

A compact matrix family without an extremal norm

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Conjecture 00000008483

Result. The singleton family consisting of a nontrivial unipotent Jordan matrix is compact and has joint spectral radius one, but admits no extremal norm. Consequently it admits no Barabanov norm.

Scope and definitions. Both source languages assert existence of Barabanov norms for compact matrix families without an irreducibility assumption. For a bounded family $\mathcal{K}$ of linear operators, write

$$M_n(\mathcal{K}) = \sup\{\|L_1 \cdots L_n\| : L_i \in \mathcal{K}\}, \quad \rho(\mathcal{K}) = \limsup_{n \rightarrow \infty} M_n(\mathcal{K})^{1/n},$$

using any fixed reference operator norm. An extremal norm N must satisfy

$$N(Lx) \leq \rho(\mathcal{K})N(x) \quad (L \in \mathcal{K}, x \in \mathbb{R}^2).$$

The Barabanov equality $\max_{L \in \mathcal{K}} N(Lx) = \rho(\mathcal{K})N(x)$ is stronger. We exclude even the displayed inequalities, for every norm. This does not contradict existence results with irreducibility hypotheses.

The compact family and its actual growth. Take

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathcal{K} = \{J\}.$$

This is a compact matrix family. On $\mathbb{R}^2$ use the reference norm $\|(x, y)\|_\infty = \max(|x|, |y|)$. Direct induction gives

$$J^n(x, y) = (x + ny, y) \quad (n \geq 0).$$

The triangle inequality implies $\|J^n v\|_\infty \leq (n+1)\|v\|_\infty$. At $v = (1, 1)$, equality holds, so the actual induced operator norm is

$$\|J^n\|_\infty = n + 1.$$

Every length- n word in $\mathcal{K}$ is precisely J^n . Thus the supremum over all such words is $M_n(\mathcal{K}) = n + 1$, and

$$\rho(\mathcal{K}) = \lim_{n \rightarrow \infty} (n+1)^{1/n} = 1.$$

For example, the last limit follows from $\log(n+1)/n \rightarrow 0$ and continuity of the exponential. This computes the actual joint spectral radius from products, not just an asserted eigenvalue.

Theorem 1. *There is no norm N on $\mathbb{R}^2$ such that $N(Jx) \leq N(x)$ for every x .*

Proof. Assume such a norm exists. Iterating the inequality yields $N(J^n x) \leq N(x)$ for every $n \geq 0$. Let $e_1 = (1, 0)$ and $e_2 = (0, 1)$. Since $J^n e_2 = n e_1 + e_2$, homogeneity and the triangle inequality give

$$nN(e_1) = N(J^n e_2 - e_2) \leq N(J^n e_2) + N(e_2) \leq 2N(e_2)$$

for every natural number n . But $N(e_1) > 0$ by positive definiteness. Choosing $n > 2N(e_2)/N(e_1)$ is a contradiction. $\square$

Since $\rho(\mathcal{K}) = 1$, the theorem excludes every extremal norm and hence every Barabanov norm for this compact family. The invariant line $\mathbb{R}e_1$ makes the family reducible; this is precisely why compactness alone is insufficient. The other source conjuncts about uniqueness and computational rates are unnecessary to the disproof.

Formal correspondence. The Lean project defines the actual continuous linear map $J(x, y) = (x + y, y)$ on the product normed space $\mathbb{R} \times \mathbb{R}$, whose norm is the maximum norm. It also defines the displayed two-by-two matrix and proves that its matrix-vector multiplication is exactly this map. The singleton operator family is proved compact.

The theorem `power_apply` covers every power and vector. The theorem `power_norm` proves the exact operator norm using both the universal upper bound and the unit vector $(1, 1)$ attaining it. The definition `products` ranges over all finite lists of the prescribed length with every factor in the family; `products_singleton` proves its full classification. The quantity `jointBound` is the genuine real supremum of norms of these products. Its exact value, the actual limit of its n -th roots, and the resulting `jointRadius` limsup are all proved.

An arbitrary candidate norm is represented by a Mathlib `Seminorm` with the additional positive-definiteness condition $N(x) = 0 \Rightarrow x = 0$. This includes every real norm and does not impose any extra continuity, coordinate, or reference-norm restriction. The theorem `no_nonexpansive_norm` proves the universal obstruction using its actual triangle inequality and scalar homogeneity. The final theorem `no_extremal_norm` combines it with the computed joint spectral radius.

Reproduction. The project pins Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. Run `lake build` from `lean/`. Principal theorems print their axiom dependencies. No admitted facts, additional axioms or external decision procedures are used.